

Declaring a variable

Ex. int x;

Declaring an ArrayList

ArrayList<*type*> *name of arrayList*;

Ex. This makes an arrayList of integers called "intList":

ArrayList<int> intList;

Initializing an ArrayList

name of arrayList = new ArrayList<*type*>();

Ex. This initializes an arrayList of ints called "intList":

intList= new ArrayList<int>();

Using an enhanced for loop

For (Ball aBall : ballList){

// do stuff to each ball in the list

}

ArrayList Commands

-arrayList commands. Lets assume we are working with an arrayList named "itemList"

- item = itemList.get(i): this sets the variable "item" equal to the ith thing in itemList.
- itemList.add(new Item()): this adds a new item to itemList
- itemList.size(): this returns the size of itemList.

Using a for loop

This example is a for loop that will run its code 50 times:

```
for (int i=0; i<50; i++){  
    *Code to run for each iteration*  
}
```

mousePressed function

```
void mousePressed(){  
    *code to run whenever the mouse is clicked*  
}
```

If statements

Example of using booleans:

```
if (graceIsTheBestTA == true){  
    println("Thanks Grace!!!");  
} else if (graceIsANerd == false){  
    println("Wait, Grace isn't a nerd?");  
} else {  
    println(" Grace is a nerd and not the best TA");  
}
```

Example of using Boolean expression:

```
if (x < 0){  
    println("x is small");  
}
```

Making a class

Example of making a class called Ball:

```
class Ball{  
    // variables  
  
    int x;  
  
    int y;  
  
    // constructor function  
    Ball (int tempX, int tempY){  
        x = tempX;  
        y = tempY;  
    }  
  
    // functions  
}
```

Defining a function

```
*function type* *function name*(*inputs*){  
*code to run in function*  
}
```

Example of a function that draw a red rectangle:

```
void render(){  
    fill(color(255,0,0));  
    rectangle(100,100,200,10)  
}
```

Example of a function that outputs true if x is between 100 and 200:

```
boolean isPositioned(int x){  
    if (x>100 && x<200){  
        return true;  
    } else {  
        return false; }}
```

Boolean operators

&& means "AND"

|| means "OR"

! means "NOT"

> means "greater than"

>= means "greater than or equal to"

< means "less than"

<= means "less than or equal to"

== means "equal to"

!= means "not equal to"

Other helpful functions and commands

- random(a,b): makes a random float between a and b.

- size(w,h): makes the window have a width of w and a height of h.

-background(c): makes the background have the color c

-circle(x,y,d): draw a circle at point x,y with a diameter of d.

-noFill(): makes it so that anything that is drawn after this command does not have a fill.

-fill(c): makes it so that the fill of any shape drawn after this command will have a color c

-stroke(c): makes it so that the stroke of any shape drawn after this command will have a color c.

- strokeWeight(num): makes the thickness of the stroke be equal to num.